Heterogeneous Verification for Multi-Step Decentralized Workflows

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Abstract

Financial and confidential automation relies on trusted intermediaries, imposing fees and custody risks that limit practical applications. Blockchains verify onchain operations but cannot verify automation requiring confidential execution, external data, or multi-step coordination. We propose a verification system where each operation uses the mechanism appropriate to its computation. The system combines cryptographic proofs for blockchain operations, hardware attestations for confidential execution, and stake-backed attestations for external interactions into workflow attestations. Validators broadcast attestations backed by stake, maintaining verifiability as long as honest validators control sufficient stake to make fraud economically irrational. This enables verifiable automation for financial operations, multi-party workflows, and confidential computation requiring trusted intermediaries.

1 Introduction

Blockchains establish cryptographic verification for single transaction operations, enabling trustless execution of financial transfers and smart contracts. While this works well for onchain computation, verification guarantees are lost when automation requires offchain execution. Multi-step workflows that interact with external APIs, process confidential compute, or perform nondeterministic operations cannot maintain end-to-end verifiability under current blockchain architectures. Developers face a binary choice: constrain all computation to onchain within gas limits and block times, or move computation offchain and abandon cryptographic verification. This forces workflows to trade functionality for verification or verification for functionality. Existing oracle networks provide partial, domain-specific feeds, but no general mechanism exists for verifiable multi-step automation across heterogeneous trust environments.

What is needed is a verification system that adapts to each computation's trust properties rather than imposing uniform assumptions. Cryptographic proofs verify blockchain operations, hardware attestations protect confidential execution, and stake-backed attestations cover external interactions, composing into unified workflow attestations. We propose a heterogeneous verification system in which validators attest to each workflow step using the appropriate mechanism and stake against their attestations; fraudulent attestations can be challenged via onchain fraud proofs. The system maintains verifiability provided honest validators control sufficient stake to make fraud economically irrational.

References

[1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.